

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

The estimator recovers, from dynamics alone, a physical property that sorts the GPS constellation into its hardware blocks with **zero misclassifications** — verified against an independent source, with the obvious confound measured and shown an order of magnitude too small to explain it. That is established. It is also established in the one regime where nobody needs it: large, cooperative, continuously tracked satellites whose masses are already published. The commercial question is no longer whether the capability is real. It is whether it survives short, sparse, noisy arcs — and that is now being tested with data already in hand, rather than by building the expensive thing first.

What was established

A single estimated parameter — an effective AC_R/m , since a cannonball fit cannot separate reflectivity from area — was fitted to 57 arcs across two constellations. Sorted by that value, 31 GPS satellites reproduce the five hardware blocks recorded in the IGS antenna file, with no misclassifications. The fit was never told that blocks exist.

evidence	result
Within-block agreement (IIF, $n = 12$)	0.29% standard deviation
Between-block separation	5–15 pooled standard deviations
Reproduction on an independent arc date (7 satellites, all 5 blocks)	0–0.5%
Block attribution source	IGS ANTEX, date-specific validity

The confound was measured, not argued away

Within each block — hardware held constant — the estimated parameter does correlate with the fit residual: $r = +0.80, +0.64, +0.56, +0.86$. Error absorption is real. But the effect size settles it: in the largest block, a $2.2\times$ swing in residual moves the estimate by 1%, against a 34% gap to the next block. The confound is an order of magnitude too small.

The check that absorbed error has no reason to pass

Multiplying each block’s estimate by its published on-orbit mass — a quantity the fit never saw — puts three GPS IIR sub-blocks within 4% of a common effective area. Those three share a spacecraft bus, and the fit was not told that either. The implied areas also order correctly against known solar-array sizes across blocks.

For absorbed model error to produce that, it would have to scale with each block’s published mass so as to land three sub-blocks on one area. No mechanism does this.

The sharpest single result is narrower and harder to explain away: two GPS III satellites — the worst-fitting block — return identical estimates to four significant figures on an independent arc, *at two different values*. Absorbed error would have to be simultaneously reproducible across days and satellite-specific. It cannot be both.

What is not claimed

Two blocks are small ($n = 5$ and $n = 2$) and carry no load. The sweep rests on one arc date, with a seven-satellite subset on a second. The GPS III block shows 6.64% internal scatter against 0.29% for the largest block, so its classification holds while its *value* does not deserve quoting; that is the weakest leg and it is under test. And all of this is GNSS.

Why that last point is the whole commercial question

GNSS satellites are large, cooperative, continuously tracked, and already characterised. Their masses are published — which is precisely why they could serve as truth here. **Nobody will pay to have an area-to-mass ratio recovered for a GPS satellite.**

The regime where this capability is worth money is the opposite in every respect: small or irregular objects, uncooperative, sparsely observed, short arcs, noisy measurements, and no published truth to check against. That is space situational awareness and debris characterisation — and it is what the Cone_HAMR and Large_Sphere_HAMR property sets in this source were built for.

So the finding is real and its commercial relevance is undetermined. Those are different statements and the gap between them is now the only thing that matters.

The decision that was deliberately not taken

The obvious next step is to build observation models, so the tool can ingest radar and optical tracks instead of published precise ephemerides. It is the largest remaining piece of work and it is the difference between a demonstration and a service.

It has been held. Its value depends entirely on a question answerable far more cheaply: does the recovery survive short, sparse, noisy arcs at all? That can be tested with data already in hand, by degrading the arcs that currently work — shortening them from 22 hours to 1, thinning and irregularising the sampling, and adding noise at decimetre, metre and tens-of-metre levels — until the block separation fails.

The deliverable is one sentence: *separation survives to X-hour arcs at Y-metre noise with Z observations, and fails beyond that*. If it survives a realistic regime, observation models are clearly worth building and will be commissioned immediately. If it collapses, the finding is real and commercially inert, and that will have cost days rather than months.

A formal uncertainty on the estimate is being added alongside it — the fit already inverts the normal matrix, so the parameter covariance is sitting there uncollected. Every figure above is currently quoted without one, and the honest breakdown point is where the formal uncertainty exceeds the between-block separation.

Where the asset stands

Revision 1	The ray-traced SRP engine — scarce, and missing its fuel
Revision 2	The validation — a credential making the SRP claim falsifiable
Revision 3	A parameter-estimating orbit determination capability
Revision 4	A robust estimator that <i>may</i> characterise objects from dynamics alone
Revision 5	It does — for large, well-tracked objects. Whether that transfers to the objects anyone would pay to characterise is the open question, and it is under test.

What still blocks a transaction

- **No licence or copyright statement anywhere in the first-party code**, which originates from UCL SGNL. Still the single largest obstacle, and it needs no engineering.
- **CALCEPH is triple-licensed** and statically linked into every binary with no choice recorded. Optional; the build drops it cleanly.
- **Credentialed data access**. LEO validation is blocked on an Earthdata login. LEO is a genuinely different regime rather than a degraded version of this one, so it is the most valuable single input still outstanding.
- **No observation models and no covariance propagation**. Deliberately held, as above. If the capability transfers, these are the decisive investment.

Recommendation

Unchanged in shape, sharper in content: decide nothing commercially until the degradation study reports. It is days of work and it determines which business this is.

1. **Finish the degradation study.** *In progress.*
2. **If the capability survives a realistic regime:** build observation models and covariance. That converts a tool that fits published ephemerides into one that characterises objects from real tracking data, and the market is SSA, not orbit determination. Settle the IP position before committing the effort, not after.
3. **If it does not:** stop, and publish. A validated estimator that recovers spacecraft hardware classes from dynamics alone is a genuinely interesting result on its own terms, and the tree is a well-documented body of work with an honest accuracy figure attached. That is a respectable place to finish and should not be treated as a failure.
4. **Settle the IP position in parallel, either way.** It gates every outcome.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026 by the `odl` maintainer session, reviewed and spot-verified here. The competitive and valuation sections are judgement. Revision 5 supersedes revision 4, which reported the hardware-block finding as untested; it has since been tested and the confound quantified. Per-block values for GPS III are withheld pending a further diagnostic.